# ESTONIAN HOUSING AFFORDABILITY DASHBOARD EXPLAINED

## PROJECT OVERVIEW

This interactive dashboard shows how affordable buying a home in Estonia is: it compares housing (apartment) prices per m² with average salaries by county over more than 20 years (2005–2026). The year 2026 covers the first two quarters only.

Data source:

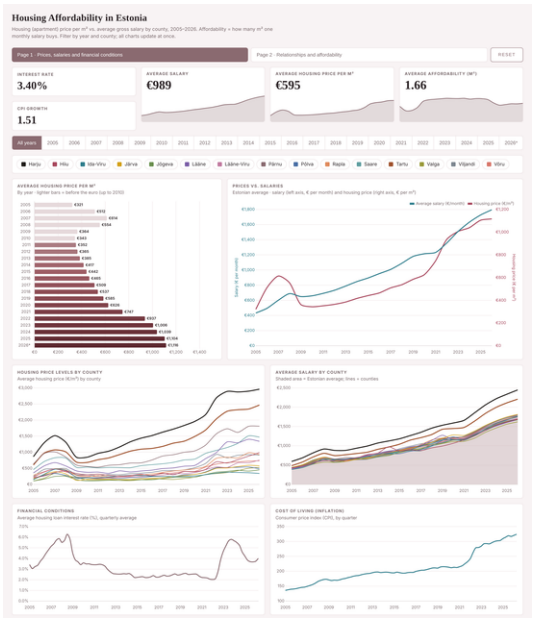
- Housing prices: Maa-amet
- Salary statistics (PA21 (2005–2017), PA004 (2018–2020), PA117 (2021–2026)): Statistikaamet
- Housing loan interest rates: Eesti Pank
- CPI (IA002): Statistikaamet

## DASHBOARD

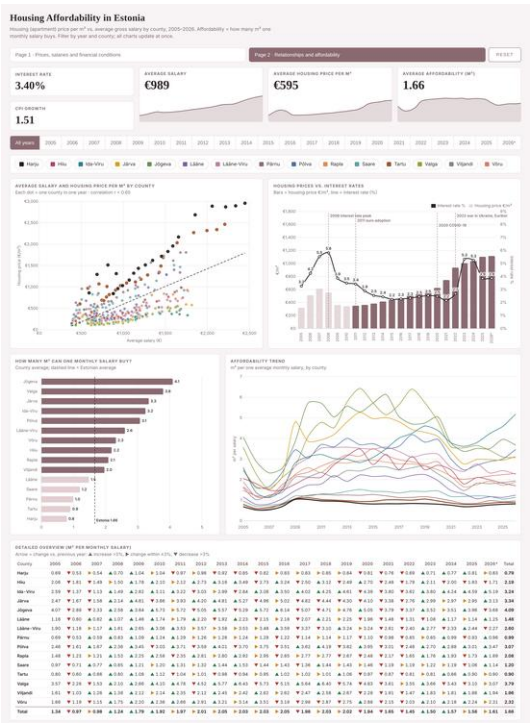
The dashboard has two pages that share one set of filters. The key indicators, the year filter and the county filter sit at the top of both pages. Page 1 covers prices, salaries and financial conditions; page 2 covers the relationships between them and affordability. Each chart answers a separate question, and the year filter, county filter and “RESET” button update all of them at once.

The dashboard answers three questions:

- How have the average housing price per m² and the average salary changed over the years?
- How many m² of housing can be bought with one monthly salary, and how does this differ by county?
- How are housing prices related to interest rates?



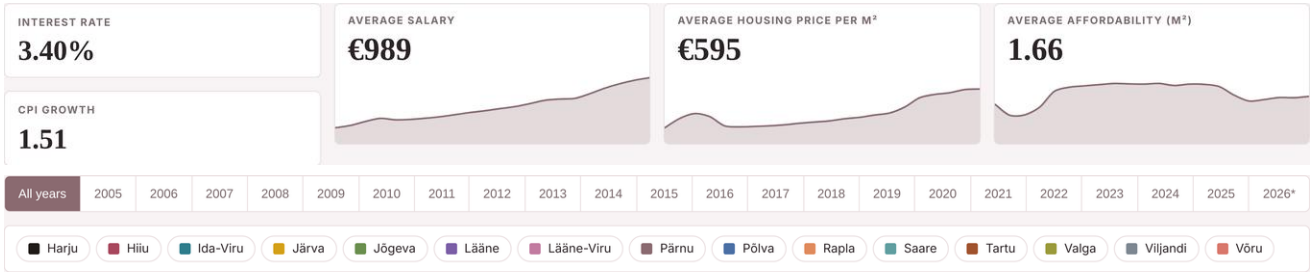

*Page 1: prices, salaries and financial conditions*




*Page 2: relationships and affordability*

## KEY INDICATORS

The top row gives a quick picture of where the market stands. The three large cards also include a small trend chart. The values below are shown with all years and all counties selected.

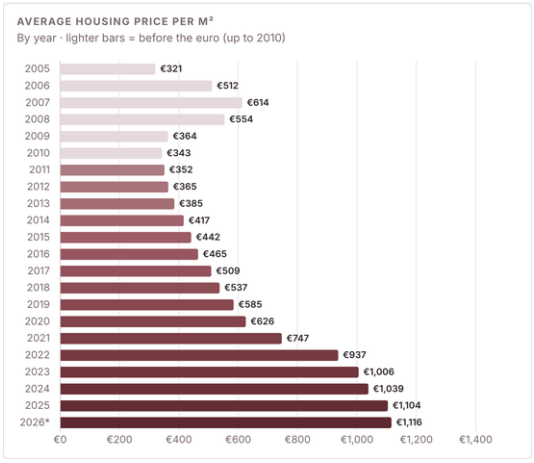



**Filters.** The year strip filters one or several years ("All years" clears the selection). The county chips filter one or several counties and use the same colours as the lines and dots in the charts, so they also work as a legend. The "RESET" button clears both filters.

| | |
|---|---|
| Interest rate | The cost of borrowing: average housing loan interest rate over the selected period (3.40% for all years) |
| CPI growth | Growth of the consumer price index, i.e. overall price inflation: average CPI of the selected period divided by the average CPI of its first year (1.51 for all years) |
| Average salary | Average monthly gross salary across the selected years and counties (€989) |
| Average housing price per m² | Average housing price per square metre (€595) |
| Average affordability | How many m² one average monthly salary buys (1.66 m²) |

# VISUALS EXPLAINED

## AVERAGE HOUSING PRICE PER M²:

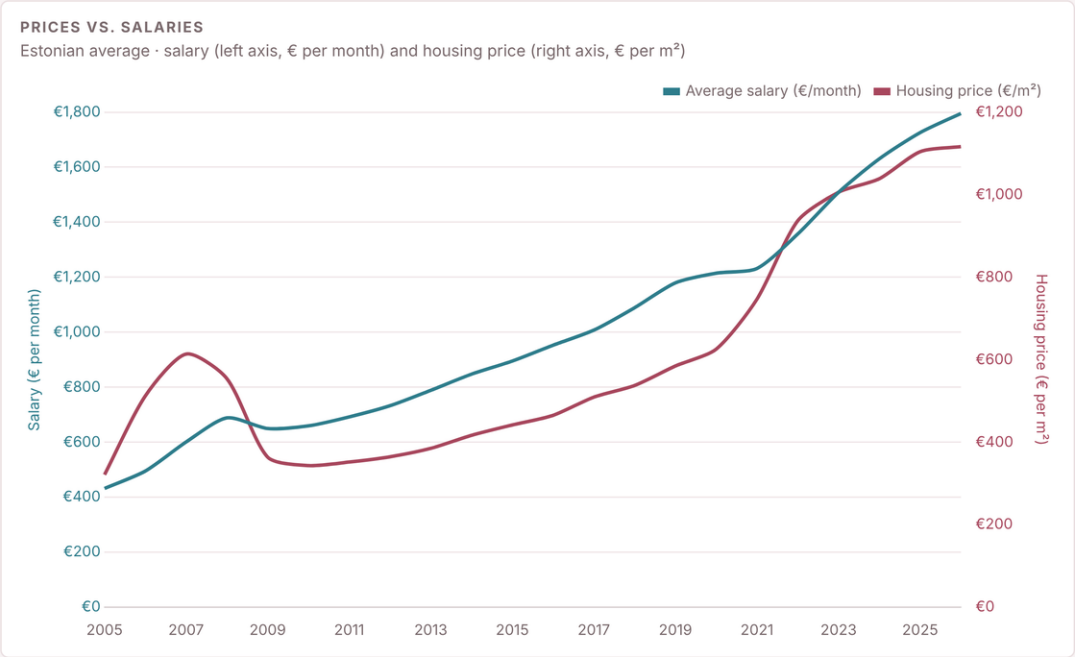



**Axes:** The Y-axis shows years and the X-axis shows the price in euros.

**Trend and visuals:** The bars get progressively darker from top to bottom, emphasising the price growth of recent years. The lighter bars (up to 2010) are the years before the euro. The year 2026 is marked with an asterisk because it covers Q1–Q2 only.

**Data:** The chart shows the peak of the 2007 boom (€614), the drop that followed during the economic crisis (the 2010 low of €343, 44% below 2007), slow growth through the 2010s and a sharp rise since 2020: from €626 to €1,116 in 2026, an increase of 78%. Over the whole period the price grew by 247.6%, from €321 in 2005.

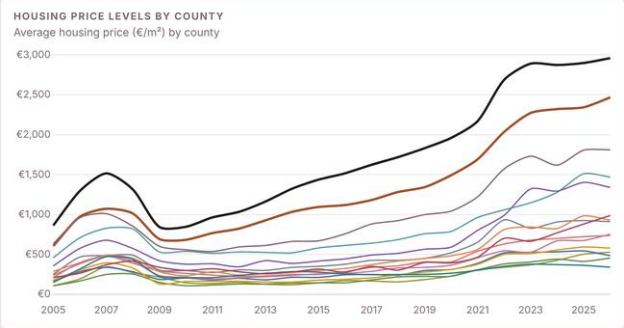
## PRICES VS. SALARIES:




Estonian average monthly salary (left axis, blue) and housing price per m² (right axis, red) on the same timeline, shown in real euro values. Both axes start at zero. Because the two lines use different axes, the points where they cross have no meaning; the chart is for comparing the trends.

- **Salary:** rises almost without interruption, from €432 in 2005 to €1,795 in 2026 (4.2 times). The only dip is in 2009 (€688 to €649).
- **Price:** moves in cycles: boom in 2005–2007 (€321 to €614), crash to €343 in 2010, slow growth in the 2010s and a sharp rise after 2020. Over the whole period it grew 3.5 times.
- **Since 2020:** prices have grown faster than salaries (+78% vs +48%).

## LINE CHARTS:

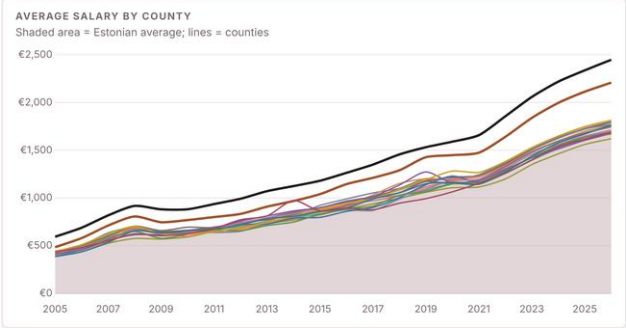



### HOUSING PRICE LEVELS BY COUNTY

Average housing price per m² for each county over time. Each line is one county; the colour chips below the filters list them. Harju (black) and Tartu (brown) are drawn thicker.

**Trend:** Prices rise to a first peak around 2007, fall during the economic crisis, then grow slowly through the 2010s. After about 2020 they climb much more steeply.

**Gap between counties:** In 2026 Harju averages €2,959/m² and Tartu €2,466/m², Pärnu €1,810, while ten counties are below €1,000. Harju is 8.7 times as expensive as Ida-Viru (€340).

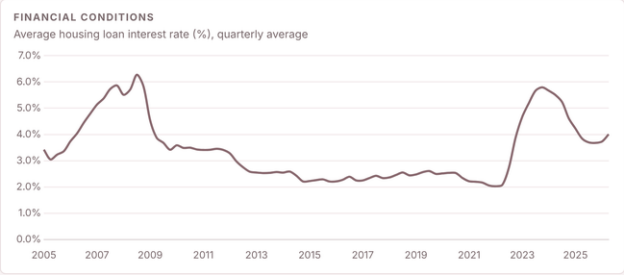



### AVERAGE SALARY BY COUNTY

Average monthly salary by county over the same period. The shaded area shows the Estonian average. Placed next to the price chart, it shows that prices rose more sharply than salaries after 2020.

**Trend:** The lines rise much more smoothly than the price lines. Salaries grow almost without interruption, most steeply in 2022–2024.

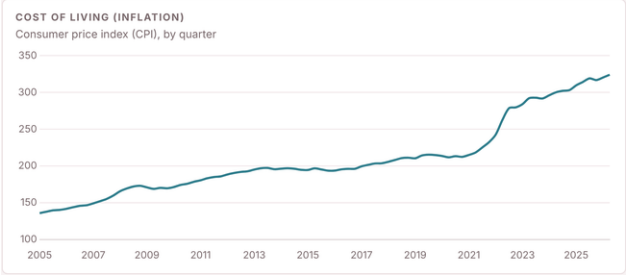
**Gap between counties:** Harju (€2,447) and Tartu (€2,208) again separate from the pack in 2026, but the spread is far narrower than for prices: the other 13 counties lie between €1,620 and €1,815, and Harju's salary is 1.5 times that of Valga, the lowest.




### FINANCIAL CONDITIONS (INTEREST RATE)

Single line showing the average housing loan interest rate (EUR loans) by quarter.

**Trend:** Two clear peaks: 2008 (5.83% annual average, quarterly maximum 6.27%) and 2023–2024 (5.33% and 5.25%). Between 2012 and 2021 the annual average stayed between 2.16% and 2.89%. In 2025–2026 it eased to about 3.9%. Rates more than doubled between 2021 (2.16%) and 2023 (5.33%), yet prices kept rising (+35%).




### COST OF LIVING (INFLATION)

Single line showing the consumer price index (CPI) by quarter.

**Trend:** Steady growth over 2005–2020 (+54% in annual average), then a clear step up in 2021–2023 (+30%), matching the inflation spike in Estonia. The latest value (Q2 2026) is 2.4 times the level of Q1 2005. It gives context for the price and salary charts: part of the rise in prices and salaries is general inflation, not real growth.

## AVERAGE SALARY AND HOUSING PRICE PER M² BY COUNTY:




Each dot on the chart represents the data of one specific county in one specific year (15 counties × 22 years = 330 dots). The dots are coloured by county, as in the colour chips below the filters; Harju is black and Tartu is brown.

- **X-axis (horizontal):** Average salary in euros
- **Y-axis (vertical):** Housing price per square metre in euros

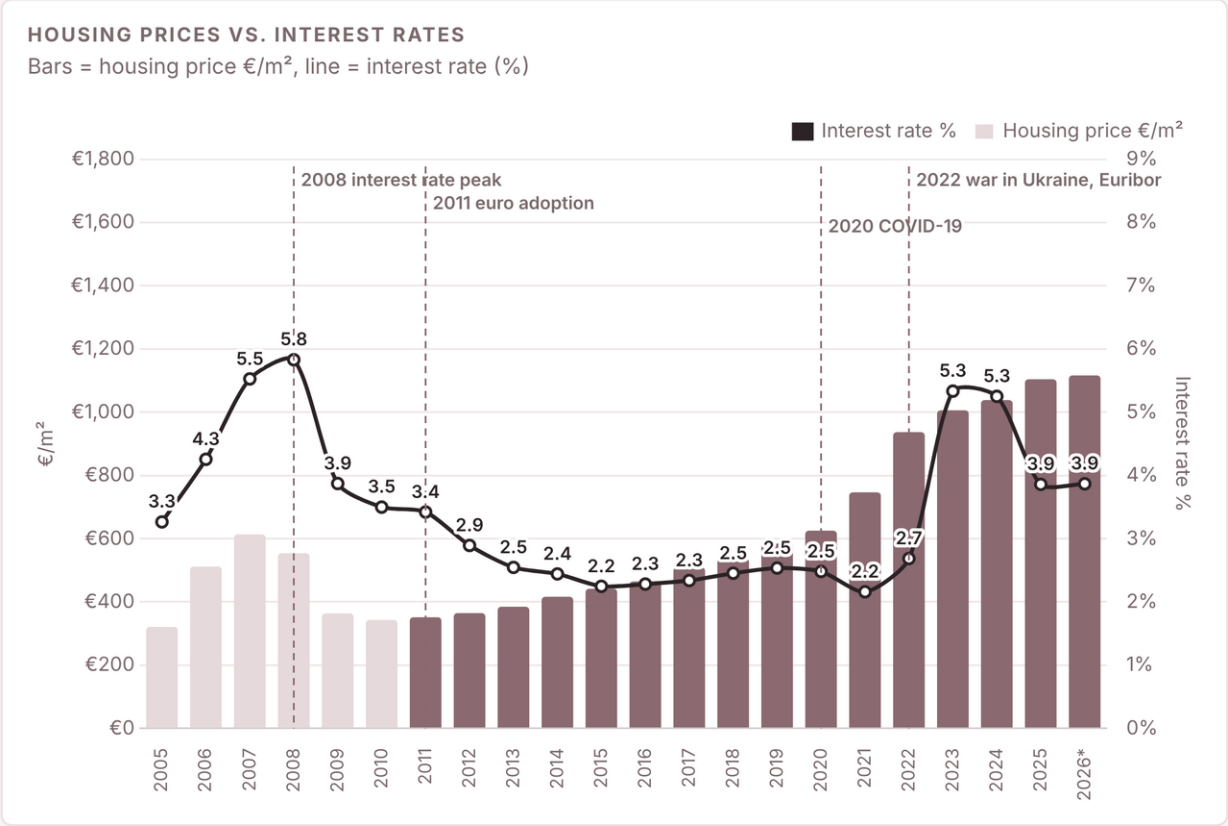
**TYPE OF RELATIONSHIP: POSITIVE CORRELATION**

The dashed diagonal line is a trend line fitted to all dots. It rises from the bottom left to the top right, so the relationship is positive: counties and years with higher salaries tend to have higher housing prices. The correlation coefficient is 0.65, a clear but not perfect relationship. It does not prove cause and effect, because price also depends on location, demand and financing conditions.

**TWO DIFFERENT GROUPS OF DATA (THE "SCISSORS" ON THE CHART):**

- **Upper right group (dark dots):** Harju and Tartu. In 2026 Harju reaches a salary of €2,447 and a price of €2,959/m²; Tartu €2,208 and €2,466/m². In these two counties each extra €100 of salary goes with about €123/m² higher prices.
- **Lower dense group (coloured dots):** The other 13 counties sit in the lower part of the chart: salaries up to €1,815 and 94% of dots below €1,000/m². Here each extra €100 of salary goes with only about €47/m². The main exception is Pärnu (€1,571–€1,810/m² in 2022–2026). Outside Harju and Tartu, salary growth does not lead to such steep price increases.

## HOUSING PRICES VS. INTEREST RATES




This chart puts 22 years of Estonian housing prices next to housing loan interest rates, to show how borrowing costs and prices have moved. The bars show the average price per m² and the line shows the average interest rate. The bars before the euro (up to 2010) are lighter. The dashed vertical lines mark key events.

### KEY EVENTS MARKED ON THE CHART

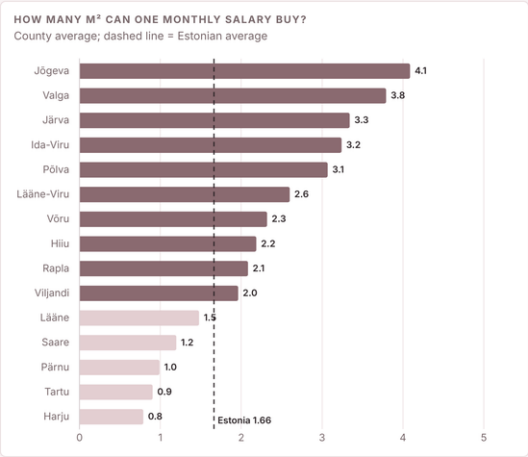
- **2008, housing boom and interest rate peak:** Prices peak at €614/m² in 2007; the interest rate peaks at 5.8% in 2008, when prices have already started falling (€554).
- **2011, switch to the euro:** After the crisis prices bottom out at €343 in 2010. From there the market recovers steadily, with interest rates falling below 3% by 2012.
- **2020, start of COVID-19:** Prices reach €626/m², finally passing the 2007 peak of €614, after 13 years. In 2021 the interest rate is at its lowest annual level (2.16%) and prices rise 19% in a year, followed by +25% in 2022.
- **2022, war in Ukraine and Euribor surge:** The interest rate jumps from 2.7% (2022) to 5.3% (2023), staying at 5.3% in 2024 (5.25%), the highest level since the 2008 peak.

### KEY OBSERVATIONS

- **Historical growth:** The average housing price per m² has grown by 247.6%, from the 2005 low (€321) to €1,116 in 2026 (Q1–Q2). For comparison, salaries grew 316% and the CPI 138.5% over the same period, so the growth is partly general inflation and rising living standards.
- **Movement in the same direction:** In 2007–2008 both the price and the interest rate peaked. In 2022–2023 the interest rate more than doubled while prices, fuelled by inflation, kept rising (to €1,006 in 2023).
- **The "scissors" in 2025–2026:** The interest rate eased from 5.3% to about 3.9%, while the price kept growing to €1,104 (2025) and €1,116 (2026*), the highest values in the series. Prices and interest rates therefore do not move one-to-one: salary growth (+48% since 2020), inflation and supply matter as well as the cost of borrowing.

## HOW MANY M² OF HOUSING CAN ONE MONTHLY SALARY BUY?

Affordability = average salary / average housing price per m².



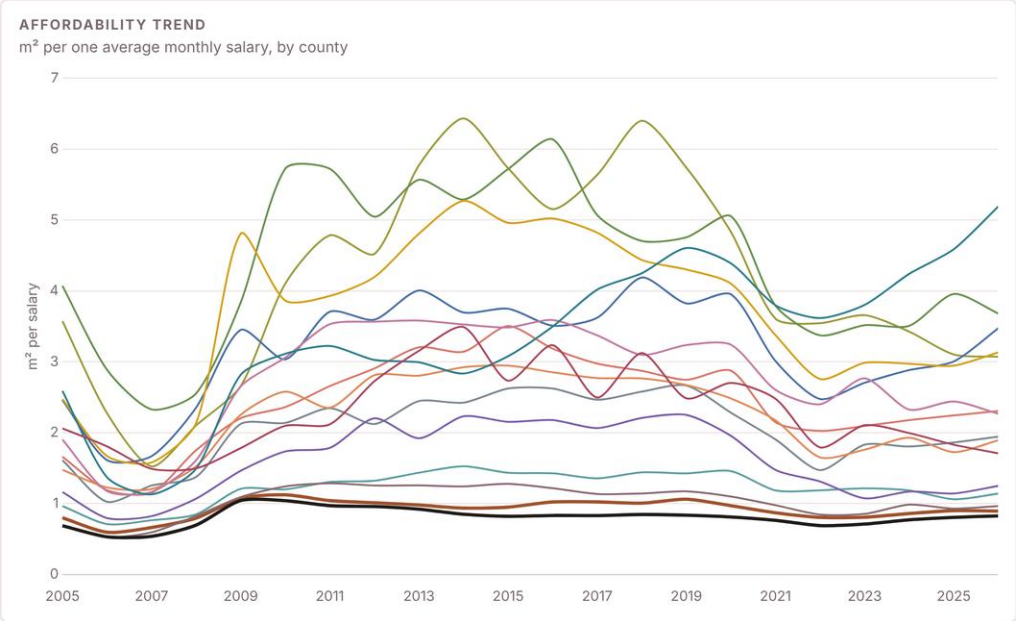

The horizontal bar chart shows overall average affordability by county across the whole period. The dashed vertical line marks the Estonian average (1.66 m²). Counties at or above the average are shown in the darker colour, those below it in the lighter colour.

**Affordability peaks (the longest bars):** Affordability is highest in Jõgeva County (4.1 m²), Valga (3.8 m²), Järva (3.3 m²), Ida-Viru (3.2 m²) and Põlva (3.1 m²). Housing there is cheap compared with local salaries.

**Affordability lows (the shortest bars):** Affordability is lowest in Harju County (0.8 m²), Tartu (0.9 m²) and Pärnu (1.0 m²). Although salaries are highest in Harju and Tartu, housing there is much harder for residents to afford than anywhere else in Estonia: one monthly salary buys less than one square metre.

*Small counties such as Hiiu have few transactions, so their figures are less stable.*

### 20-YEAR AFFORDABILITY TREND




This line chart shows how the same affordability ratio has changed over time, by county. The Estonian average (all counties) shows three eras:

- **2006–2007 (the low point, i.e. the housing boom):** The lines collapse towards the bottom of the chart. The Estonian average is 0.97 m² per salary in 2006 and 0.98 in 2007; in Harju it falls to 0.53.
- **2009–2020 (a decade of high affordability):** After the bubble burst, prices fell and a few years later salaries started to grow again. The Estonian average jumps to 1.79 in 2009 and stays between 1.9 and 2.05 until 2020, peaking in 2013 and 2016 (2.05). In some counties one salary bought more than 6 m² (Valga 6.4 in 2014, Jõgeva 6.1 in 2016).
- **2021–2026 (a new decline and partial recovery):** The lines turn clearly downwards. The Estonian average falls from 2.05 (2016) to 1.45 in 2022 (−29%), as post-pandemic price growth and the rise in interest rates took effect. It has since recovered slightly to 1.61 in 2026*, because salaries (+32% since 2022) grew faster than prices (+19%). It remains well above the 2006–2007 low.

## DETAILED OVERVIEW WITH INDICATORS

| County | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Harju | 0.69 | ▼ 0.53 | ► 0.54 | ▲ 0.70 | ▲ 1.04 | ► 1.04 | ▼ 0.97 | ► 0.96 | ▼ 0.92 | ▼ 0.85 | ▼ 0.82 | **0.79** |
| Hiiu | 2.06 | ▼ 1.81 | ▼ 1.49 | ► 1.50 | ▲ 1.78 | ▲ 2.10 | ► 2.12 | ▲ 2.73 | ▲ 3.16 | ▲ 3.49 | ▼ 2.73 | **2.19** |
| Ida-Viru | 2.59 | ▼ 1.37 | ▼ 1.13 | ▲ 1.49 | ▲ 2.82 | ▲ 3.11 | ▲ 3.22 | ▼ 3.03 | ► 2.99 | ▼ 2.84 | ▲ 3.08 | **3.24** |
| Järva | 2.47 | ▼ 1.67 | ▼ 1.58 | ▲ 2.14 | ▲ 4.81 | ▼ 3.86 | ► 3.93 | ▲ 4.20 | ▲ 4.81 | ▲ 5.27 | ▼ 4.96 | **3.34** |
| Jõgeva | 4.07 | ▼ 2.89 | ▼ 2.33 | ▲ 2.56 | ▲ 3.84 | ▲ 5.73 | ► 5.72 | ▼ 5.05 | ▲ 5.57 | ▼ 5.29 | ▲ 5.72 | **4.09** |
| Lääne | 1.16 | ▼ 0.80 | ▲ 0.82 | ▲ 1.07 | ▲ 1.46 | ▲ 1.74 | ► 1.79 | ▲ 2.20 | ▼ 1.92 | ▲ 2.23 | ▼ 2.15 | **1.48** |
| Lääne-Viru | 1.90 | ▼ 1.18 | ► 1.17 | ▲ 1.61 | ▲ 2.65 | ▲ 3.06 | ▲ 3.53 | ► 3.57 | ► 3.58 | ► 3.53 | ► 3.48 | **2.60** |
| Pärnu | 0.69 | ▼ 0.53 | ▲ 0.59 | ▲ 0.83 | ▲ 1.09 | ▲ 1.24 | ▲ 1.29 | ► 1.26 | ► 1.26 | ► 1.24 | ► 1.28 | **0.99** |
| Põlva | 2.46 | ▼ 1.61 | ▲ 1.67 | ▲ 2.36 | ▲ 3.45 | ▼ 3.03 | ▲ 3.71 | ▼ 3.59 | ▲ 4.01 | ▼ 3.70 | ► 3.75 | **3.07** |
| Rapla | 1.48 | ▼ 1.23 | ► 1.21 | ▲ 1.53 | ▲ 2.25 | ▲ 2.58 | ▼ 2.35 | ▲ 2.81 | ► 2.80 | ▲ 2.92 | ► 2.95 | **2.08** |
| Saare | 0.97 | ▼ 0.71 | ▲ 0.77 | ▲ 0.85 | ▲ 1.21 | ► 1.20 | ▲ 1.31 | ► 1.32 | ▲ 1.44 | ▲ 1.53 | ▼ 1.44 | **1.20** |
| Tartu | 0.80 | ▼ 0.60 | ▲ 0.66 | ▲ 0.80 | ▲ 1.08 | ▲ 1.12 | ▼ 1.04 | ► 1.01 | ▼ 0.98 | ▼ 0.94 | ► 0.95 | **0.90** |
| Valga | 3.57 | ▼ 2.28 | ▼ 1.53 | ▲ 2.10 | ▲ 2.66 | ▲ 4.10 | ▲ 4.78 | ▼ 4.52 | ▲ 5.77 | ▲ 6.43 | ▼ 5.73 | **3.79** |
| Viljandi | 1.61 | ▼ 1.03 | ▲ 1.26 | ▲ 1.38 | ▲ 2.12 | ► 2.14 | ▲ 2.35 | ▼ 2.12 | ▲ 2.45 | ► 2.42 | ▲ 2.62 | **1.96** |
| Võru | 1.66 | ▼ 1.19 | ▼ 1.15 | ▲ 1.75 | ▲ 2.20 | ▲ 2.36 | ▲ 2.66 | ▲ 2.91 | ▲ 3.21 | ► 3.14 | ▲ 3.51 | **2.32** |
| **Total** | **1.34** | **▼ 0.97** | **► 0.98** | **▲ 1.24** | **▲ 1.79** | **▲ 1.92** | **► 1.97** | **► 2.01** | **► 2.05** | **► 2.03** | **► 2.03** | **1.66** |

| County | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026* | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Harju | ► 0.83 | ► 0.83 | ► 0.85 | ► 0.84 | ▼ 0.81 | ▼ 0.76 | ▼ 0.69 | ▲ 0.71 | ▲ 0.77 | ▲ 0.81 | ► 0.83 | **0.79** |
| Hiiu | ▲ 3.24 | ▼ 2.50 | ▲ 3.12 | ▼ 2.49 | ▲ 2.70 | ▼ 2.48 | ▼ 1.79 | ▲ 2.11 | ▼ 2.00 | ▼ 1.83 | ▼ 1.71 | **2.19** |
| Ida-Viru | ▲ 3.50 | ▲ 4.02 | ▲ 4.25 | ▲ 4.61 | ▼ 4.39 | ▼ 3.80 | ▼ 3.62 | ▲ 3.80 | ▲ 4.24 | ▲ 4.59 | ▲ 5.19 | **3.24** |
| Järva | ► 5.02 | ▼ 4.82 | ▼ 4.44 | ▼ 4.30 | ▼ 4.10 | ▼ 3.38 | ▼ 2.76 | ▲ 2.99 | ► 2.97 | ► 2.95 | ▲ 3.13 | **3.34** |
| Jõgeva | ▲ 6.14 | ▼ 5.07 | ▼ 4.71 | ► 4.76 | ▲ 5.05 | ▼ 3.79 | ▼ 3.37 | ▲ 3.52 | ► 3.51 | ▲ 3.96 | ▼ 3.68 | **4.09** |
| Lääne | ► 2.18 | ▼ 2.07 | ▲ 2.21 | ► 2.25 | ▼ 1.96 | ▼ 1.48 | ▼ 1.31 | ▼ 1.08 | ▲ 1.17 | ► 1.14 | ▲ 1.25 | **1.48** |
| Lääne-Viru | ▲ 3.59 | ▼ 3.37 | ▼ 3.10 | ▲ 3.24 | ► 3.24 | ▼ 2.61 | ▼ 2.40 | ▲ 2.77 | ▼ 2.33 | ▲ 2.44 | ▼ 2.27 | **2.60** |
| Pärnu | ▼ 1.22 | ▼ 1.14 | ► 1.14 | ► 1.17 | ▼ 1.10 | ▼ 0.98 | ▼ 0.85 | ► 0.85 | ▲ 0.99 | ▼ 0.93 | ▲ 0.96 | **0.99** |
| Põlva | ▼ 3.51 | ▲ 3.62 | ▲ 4.19 | ▼ 3.82 | ▲ 3.95 | ▼ 3.01 | ▼ 2.48 | ▲ 2.70 | ▲ 2.88 | ▲ 3.01 | ▲ 3.47 | **3.07** |
| Rapla | ▼ 2.85 | ► 2.77 | ► 2.77 | ▼ 2.67 | ▼ 2.48 | ▼ 2.17 | ▼ 1.65 | ▲ 1.76 | ▲ 1.93 | ▼ 1.73 | ▲ 1.89 | **2.08** |
| Saare | ► 1.43 | ▼ 1.36 | ▲ 1.44 | ► 1.43 | ► 1.46 | ▼ 1.19 | ► 1.19 | ► 1.22 | ► 1.19 | ▼ 1.06 | ▲ 1.14 | **1.20** |
| Tartu | ▲ 1.02 | ► 1.02 | ► 1.01 | ▲ 1.06 | ▼ 0.97 | ▼ 0.87 | ▼ 0.81 | ► 0.81 | ▲ 0.86 | ▲ 0.90 | ► 0.90 | **0.90** |
| Valga | ▼ 5.15 | ▲ 5.64 | ▲ 6.40 | ▼ 5.74 | ▼ 4.83 | ▼ 3.61 | ► 3.55 | ▲ 3.66 | ▼ 3.43 | ▼ 3.10 | ► 3.07 | **3.79** |
| Viljandi | ► 2.62 | ▼ 2.47 | ▲ 2.58 | ▲ 2.67 | ▼ 2.28 | ▼ 1.91 | ▼ 1.47 | ▲ 1.83 | ► 1.81 | ▲ 1.86 | ▲ 1.94 | **1.96** |
| Võru | ▼ 3.19 | ▼ 2.98 | ▼ 2.87 | ▼ 2.75 | ▲ 2.88 | ▼ 2.15 | ▼ 2.03 | ▲ 2.10 | ▲ 2.18 | ▲ 2.24 | ► 2.31 | **2.32** |
| **Total** | **► 2.05** | **▼ 1.98** | **► 2.03** | **► 2.02** | **▼ 1.94** | **▼ 1.65** | **▼ 1.45** | **▲ 1.50** | **▲ 1.57** | **► 1.56** | **► 1.61** | **1.66** |

*The matrix is a single table in the dashboard; it is split into two parts here for readability. Values are m² per monthly salary.*

The coloured arrows (KPI indicators) show the change compared with the previous year: ▲ green = increase of more than 3%, ► amber = change within ±3%, ▼ red = decrease of more than 3%.

- **2006, a sea of red:** In 2006 all 15 counties are ▼. In Harju County, for example, affordability fell from 0.69 to 0.53 m² per salary, as prices rose 59% in a year and salaries only 15%.
- **The green turn of 2008–2009:** Right after the crash, affordability jumps up: 14 of 15 counties are ▲ in 2008 and all 15 in 2009 (in Harju County 0.54 → 0.70 → 1.04 m²), because prices fell faster than salaries (−41% vs +8% between 2007 and 2009). Harju is at or above 1 m² per salary only in 2009–2010 (1.04).
- **The sharp fall of 2021–2022:** In 2021 all 15 counties are ▼, and in 2022 13 of 15. Harju falls from 0.81 (2020) to 0.69 (2022) and Tartu from 0.97 to 0.81.
- **Recovery since 2023:** Harju is ▲ in 2023–2025 (0.71, 0.77, 0.81) and Tartu in 2024–2025 (0.86, 0.90), but both remain below their 2009–2010 levels (1.04 and 1.12).

## KEY OBSERVATIONS

These visuals support the conclusion that growth in gross salaries alone does not guarantee affordability in the housing market. Although salaries have multiplied 4.2 times over 21 years, the “red arrows” of 2006 and 2021–2022 show that fast price growth, inflation and rising interest rates can lower residents’ affordability within a year or two, especially in Harju and Tartu counties.

- Salaries have grown steadily (€432 → €1,795), while housing prices and affordability have moved in cycles.
- The price per m² peaked early in the period (€614 in 2007), fell 44% by 2010, and has since grown again, reaching €1,116 in 2026*.
- Interest rates have made two clear peaks: 5.8% in 2008 and about 5.3% in 2023–2024.
- Affordability rose strongly after 2008, stayed around 2 m² per salary for a decade, fell to 1.45 in 2022 and has partly recovered (1.61).
- There are large differences between counties: one salary buys 4.1 m² in Jõgeva but 0.8 m² in Harju, and Harju prices are 8.7 times those of Ida-Viru (2026).

## NOTES AND LIMITATIONS

- 2026 covers Q1–Q2 only (marked with an asterisk).
- Prices and salaries are nominal, i.e. not adjusted for inflation.
- Affordability is the ratio of the average salary to the average price per m²; county prices are simple averages, not weighted by the number of transactions.

## TOOLS AND SKILLS

- The dashboard is an interactive two-page report with a year filter, county filter and “RESET” button; it was designed in Power BI.
- Choice of visuals: bar, line, area, scatter and combo charts, plus a matrix
- Conditional formatting: colour markers (arrows) in the table
- Measures (DAX): affordability calculated as the ratio of salary to price per m²; CPI growth; change compared with the previous year
- Data preparation with Python and DBeaver (SQL), star schema model
- The web version were made with the help of Claude, built from the same data.